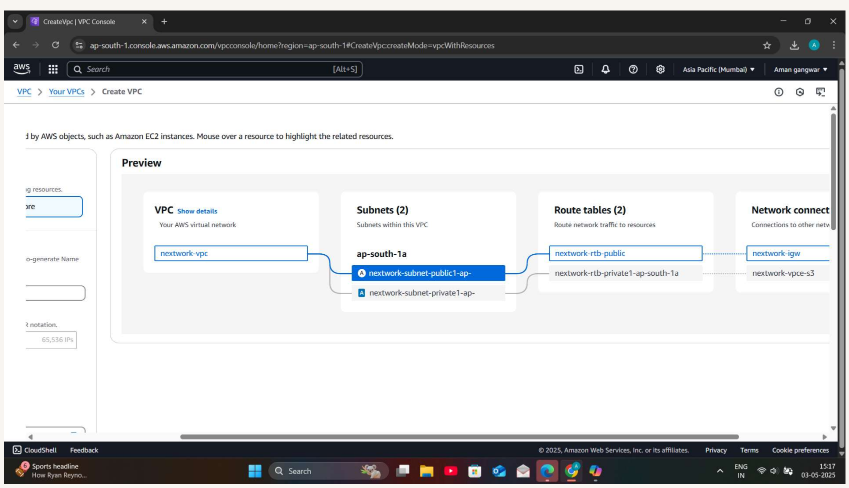
nextwork.org

Launching VPC Resources

Aman gangwar



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC (Virtual Private Cloud) is a service that allows you to create a logically isolated network within AWS, similar to a traditional data center network but with the flexibility and scalability of the cloud.

How I used Amazon VPC in this project

I used Amazon VPC to create my own VPC with different resources/components(e.g. security group ,network ACLs),private resources(e.g. a private subnet) and EC2 instances in both my public and private subnets.

One thing I didn't expect in this project was...

I didn't expect the resource map to be a visual and interactive! Such a convenient and speedy way to set up an entire VPC.

This project took me...

This project took me 2.5 hours to complete.

Setting Up Direct VM Access

Directly accessing a VM means "logging into" the EC2 instance (instead of just managing it a higher level with the AWS Management Console).This include operations like installing software and challenging my EC2 instance configurations.

SSH is a key method for directly accessing a VM

SSH traffic means Secure Shell (SSH) connections used to remotely access and manage cloud-based Linux instances. SSH allows users to securely connect to their Amazon EC2 instances from their local machines.

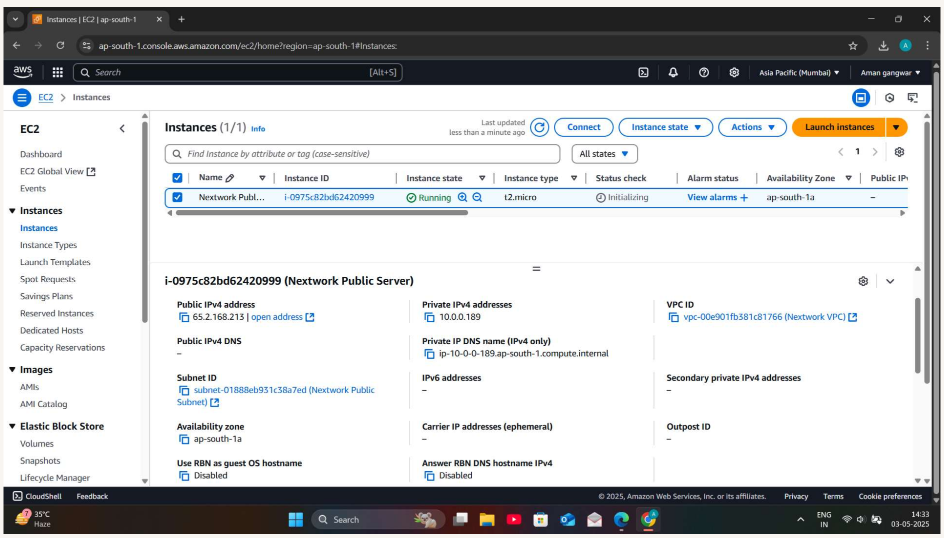
To enable direct access, I set up key pairs

Key pair are tools that help developers/engineers authenticate themselves when trying to direct access to a virtual machine .Key pair works by having two private keys- a private key for VM, and a matching private key for the resource/user

A private key's file format was .pem ,which is a widely accepted file format that most servers will be able to read/use.

Launching a public server

I had to change my EC2 instance's networking settings by changing the VPC and the subnet my EC2 instance was going to be launched in! I updated both to my Nextwork VPC and my Public Subnet respectively.I also used my existing public security group.



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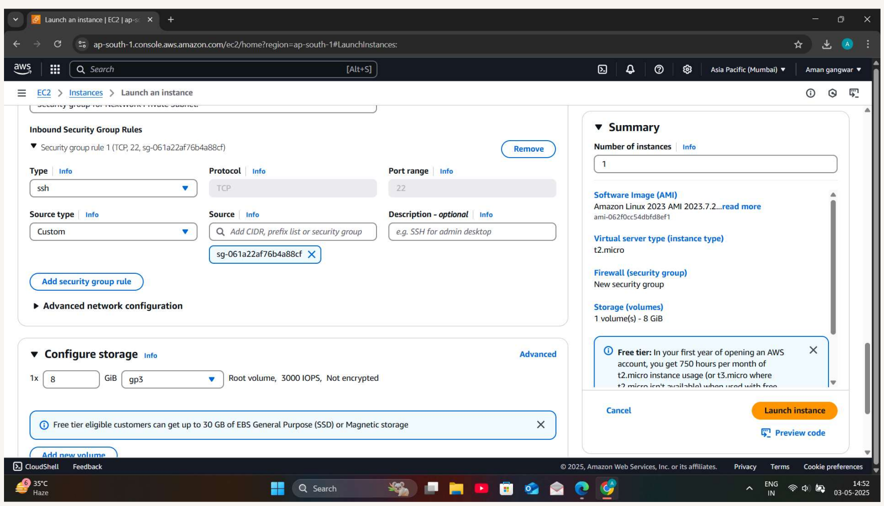
Launching a private server

My private server has its own dedicated security group because the public security group allows in ALL HTTP traffic-which would leave our private server much more vulnerable to security attacks/risks.

My private server's security group's source is my Public Security Group,which means only SSH traffic coming from resources associated with that security group would be allowed

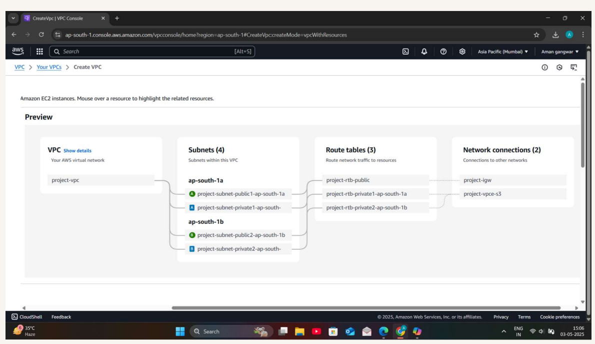


Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, I used the 'VPC and more option ,which gives me a VPC resource map to use when creating the VPC and all of its components e.g. security groups,route tables,internet gateways.

A VPC resource map is a visual representation of all the resources within your Amazon Virtual Private Cloud (VPC). It helps you understand the architecture of your VPC, including subnets, route tables, internet gateways, NAT gateways, and more.

My new VPC has a CIDR block of 10.0.0.0/16.It is possible for my new VPC to have the same IPv4 CIDR block as my existing NextWork VPC because AWS treats each VPC as an independent network, meaning they do not automatically communicate with each other



Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options either none or one in each AZ for my VPC.This was because it's best practice to have atleast 1 subnet/AZ.

The set up page also offered to create NAT gateways, which are Network Address Translation (NAT) services that allow instances in a private subnet to access the internet or other AWS services without exposing their private IP addresses